

## CODEFEST 9

- 1) Identify the dominant kernel in your accelerator (e.g., matrix multiply, convolution, dot product). State its dimensions and data types precisely.

**Dominant kernel:** Multihead Self Attention module from the Conformer algorithm - Matrix Multiple (GEMM).

**Data Type:** BF16

**Dimensions:** Tile dimension = 64

Number of heads in multihead self attention module = 8

Model dimension = 512

Head dimension = 64

- 2) Count total FLOPs (or MACs×2) for one invocation of the kernel at your design's operating point.

Calculating the FLOPs across 5 stages:

**Stage 1:** For Q, K, V projections

$$\begin{aligned}\text{Attention projections (in bytes)} &= 3 * (2 * B * T * d\_model * d\_head) \\ &= 3 * (2 * 1 * 64 * 512 * 64) \\ &= 12,582,912\end{aligned}$$

**Stage 2:**  $QK^T$

$$\begin{aligned}QK^T \text{ score} &= 2 * B * h\_head * T * (d/h) * T \\ &= 2 * 1 * 8 * 64 * 64 * 64 \\ &= 4,194,304\end{aligned}$$

**Stage 3:** Softmax

$$\begin{aligned}\text{Softmax score} &= 2 * B * h * T * T \\ &= 2 * 1 * 8 * 64 * 64 \\ &= 65,536\end{aligned}$$

**Stage 4:** Scores x V

$$\begin{aligned}\text{score} &= 2 * B * h * T * T * (d/h) \\ &= 2 * 1 * 8 * 64 * 64 * 64 \\ &= 4,194,304\end{aligned}$$

**Stage 5:** Output projections

$$\begin{aligned}\text{output projections} &= 2 * B * T * d\_model * d\_model \\ &= 2 * 1 * 64 * 512 * 512 \\ &= 33,554,432\end{aligned}$$

$$\begin{aligned}
\text{Total FLOPs} &= \text{Stage 1} + \text{Stage 2} + \text{Stage 3} + \text{Stage 4} + \text{Stage 5} \\
&= 12,582,912 + 4,194,304 + 65,536 + 4,194,304 + 33,554,432 \\
&= 54,591,488 \\
&= 54.6 \text{ M}
\end{aligned}$$

- 3) Count total bytes transferred from off-chip memory assuming no data reuse (lower bound on AI).

Count total bytes transferred assuming perfect on-chip data reuse for weights (upper bound on AI).

Both calculations must show the formula and values. If your kernel does not fit the GEMM-style weight-reuse pattern (e.g., depthwise convolution, embedding lookup, streaming sparse MVM, event-driven sparse compute), state the reuse pattern that actually applies to your kernel and compute the two bounds against that pattern instead. Name the pattern explicitly.

Assuming **No Data Reuse (Lower bound)**:

|                                                |                                                                                                             |                |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------|----------------|
| <b>Stage 1</b><br>Q, K, V Projections in bytes | $= (B * T * d + 3 * d * d_h + 3 * B * T * d_h) * 2$ $= (1 * 64 * 512 + 3 * 512 * 64 + 3 * 1 * 64 * 64) * 2$ | <b>286,720</b> |
| <b>Stage 2</b><br>$QK^T$ in bytes              | $= (2 * B * h * T * d_h + B * h * T * T) * 2$ $= (2 * 1 * 8 * 64 * 64 + 1 * 8 * 64 * 64) * 2$               | <b>196,608</b> |
| <b>Stage 3</b><br>Softmax in bytes             | $= (2 * B * h * T * T) * 2$ $= (2 * 1 * 8 * 64 * 64) * 2$                                                   | <b>131,072</b> |
| <b>Stage 4</b><br>Scores * V                   | $= (B * h * T * T + 2 * B * h * T * d_h) * 2$ $= (1 * 8 * 64 * 64 + 2 * 1 * 8 * 64 * 64) * 2$               | <b>196,608</b> |
| <b>Stage 5</b><br>Output Projections           | $= (2 * B * T * d + d * d) * 2$ $= (2 * 1 * 64 * 512 + 512 * 512) * 2$                                      | <b>655,360</b> |

$$\begin{aligned}
\text{Total (bytes)} &= 286,720 + 196,608 + 131,072 + 196,608 + 655,360 \\
&= 1,466,368 \text{ bytes}
\end{aligned}$$

$$\begin{aligned}
\text{Total AI lower bound} &= 54,591,488 / 1,466,368 \\
&= 37.23 \text{ FLOPs/byte}
\end{aligned}$$

Assuming **Perfect on-chip weight (Upper bound)**:

|                                                |                                                                  |                |
|------------------------------------------------|------------------------------------------------------------------|----------------|
| <b>Stage 1</b><br>Q, K, V Projections in bytes | $=(B*T*d + 3*B*T*dh) * 2$ $=(1*64*512 + 3*1*64*64) * 2$          | <b>90,112</b>  |
| <b>Stage 2</b><br>$QK^T$ in bytes              | $= (2*B*h*T*dh + B*h*T*T) * 2$ $= (2*1*8*64*64 + 1*8*64*64) * 2$ | <b>196,608</b> |
| <b>Stage 3</b><br>Softmax in bytes             | $= (2*B*h*T*T) * 2$ $= (2*1*8*64*64) * 2$                        | <b>131,072</b> |
| <b>Stage 4</b><br>Scores * V                   | $= (B*h*T*T + 2*B*h*T*dh) * 2$ $= (1*8*64*64 + 2*1*8*64*64) * 2$ | <b>196,608</b> |
| <b>Stage 5</b><br>Output Projections           | $= (2*B*T*d) * 2$ $= (2*1*64*512) * 2$                           | <b>131,072</b> |

Total (bytes) = 90,112 + 196,608 + 131,072 + 196,608 + 131,072  
= 745,472 bytes

Total AI upper bound = 54,591,488 / 745,472  
= 73.23 FLOPs/byte